

SKILLS

Data Analysis: Data Cleaning, Data Preprocessing, Data Validation, Model Evaluation

Programming: Python, SQL, R

Libraries: NumPy, Pandas, Scikit-learn

AI & ML Frameworks: PyTorch, Keras, scikit-learn

Tools: AWS, cPanel, AlmaLinux

PROJECTS

Enhancing NPC Interaction in Adventure Game Using Deep Reinforcement Learning *Jul 2024 - Apr 2025*

- Engineered an adaptive AI game using Deep Q-Networks (DQN) and Python to drive dynamic NPC behavior in real-time Unity environments.
- Integrated Finite State Machines (FSM) with neural networks, enabling agents to autonomously optimize actions from continuous feedback.
- Analyzed and tuned algorithmic performance during training, monitoring reward functions and convergence metrics for stable agent behavior.

Plant Disease Detection Using DCNN

Jan 2024 - Jun 2024

- Engineered a Deep Convolutional Neural Network (DCNN) to automate detection and classification of plant diseases from image data.
- Applied image preprocessing and data augmentation to mitigate data imbalances and improve generalization to unseen conditions.
- Optimized network architecture through hyperparameter tuning, evaluating accuracy, precision, and loss to ensure reliable classification.

CERTIFICATION

Linux System Administration — Clado Solutions, Kerala

May 2025 - Jul 2025

- Deployed and administered cloud-based server infrastructure on AWS and AlmaLinux, ensuring optimal configuration, security, and high availability.
- Managed web hosting environments and system resource allocation using cPanel, streamlining server administration workflows.
- Investigated and resolved technical issues through mock support tickets, simulating real-world troubleshooting and root-cause analysis via server log evaluation.

Machine Learning — Keltron Knowledge Centre, Kerala

Apr 2023

- Conducted statistical analysis and exploratory data evaluation to identify trends, outliers, and underlying patterns within complex datasets.
- Built predictive pipelines using Linear Regression, Logistic Regression, and Decision Trees to forecast numerical and categorical outcomes.
- Segmented and categorized multidimensional data using K-Means Clustering, K-Nearest Neighbors (KNN), and Support Vector Machines (SVM).

RESEARCH PUBLICATION

Enhancing NPCs Using Deep Reinforcement Learning

International Journal of Scientific Research in Science and Technology (IJSRST), Vol. 12, Issue 12, 2025

EDUCATION

Viswajyothi College of Engineering and Technology, Kerala

2021 - 2025

B.Tech in Artificial Intelligence and Data Science | CGPA: 8.27

Army Public School, Mumbai

2020

Higher Secondary Education (CBSE) | 91.60%

Vimalagiri International School

2018

High School Education (ICSE) | 91.40%